

AI Fo 2025 Graded Miniproject

This document specifies the miniproject. It will be graded and it counts 40% of your AI Fo grade (the other 60% come from the written exam).

1. Important Dates:

Start: Friday, September 26th 2025

Deadline: **Sunday, November 9th 2025 (upload your report before 23:50)**

2. Problem-Based Learning and Teamwork!

- This is a "Problem-Based Learning" project. We provide only a minimum of instructions and you solve the task in a team.
- Work in teams of two people. Larger teams are not allowed (except one team of 3 people if the number of students is odd). Working alone is not allowed. Use the Discussion Forum on Moodle to form teams.
- Collaborate effectively! You learn more and you are faster when you discuss your ideas and problems.
- Both team members will get the same grade. We reserve the right to interview (randomly selected) team members in order to make sure no one gets a free ride! Note that fraud is not accepted at OST and may have severe consequences.

3. Tasks

a) Explore the *technical* possibilities of GPT Assistants.

It is not difficult develop a basic assistant. Use the playground to familiarize yourself with the openAI platform: <https://platform.openai.com/playground/assistants>

In particular, have a look at the options:

- System instructions
- Files / File Search
- Code Interpreter
- Functions

You can use the OpenAI GUI <https://platform.openai.com/assistants> to create and test an assistant and use the Responses API <https://platform.openai.com/docs/api-reference/responses> to interact with the OpenAI services

Technical Note:

In the openAI platform GUI, you can enable "Functions" and define the *interface* (but not the code) of a function. The actual implementation (and call) of a function is on the "user-side" (typically within a company, or a third party webservice). When testing it on the playground, you can simply enter a "mock result".

b) Generate and document at least 3 ideas

Spend some time on developing a good idea. Discuss the possibilities of Assistants and GPTs. Think of opportunities in your business-context or related to your hobbies and interests. Dig a bit - instead of just implementing the first trivial use-case.

Try an idea-generation technique:

<https://www.indeed.com/career-advice/career-development/idea-generation-techniques-besides-brainstorming>

Note: ideation is important! Be creative, do not just jump on the first idea. (Almost) all ideas are welcome, but we explicitly exclude the following two topics (because we have seen too many of them already, or because they are trivial)

- Dungeon and Dragons game assistant/master
- Kitchen assistant, food assistant.

c) Evaluate your ideas

Compare the different ideas. In your group, agree on different evaluation criteria and/or weighting of criteria. For example, is it important for you to create something funny, or something with high business potential, or something that is beneficial for society?

Based on your evaluation, select one idea for the next steps.

d) Implement one idea

Go to <https://platform.openai.com/playground/assistants> and implement your favourite idea. Note that implementing your idea might be trickier than you thought. While learning a new platform, it is normal to go back again to the steps a) to d) and refine the project.

Your implementation should consist of at least two parts: **an openAI Assistant and a client software** (at least a simple command line interface). You are free to include additional services if needed.

e) Evaluate your implementation

What works? What doesn't? What are your experiences with the platform? What would you implement next if you had more time?

f) Write a short report

Describe and summarize the process and results in a short report (4 to max 6 pages, plus a title page and an optional appendix). Make sure to document all of the aforementioned tasks. Make sure the report is concise and includes diagrams that explain the architecture and the application.

Use of chatGPT: it's ok to use it as copy editor. If used for ideation, evaluation or coding, then declare it. Do not let chatGPT write empty blabla paragraphs.

3.1. Minimal requirements

To obtain a **grade 4.0**, the following **minimal requirements** need to be implemented:

- Implement a small command-line application. The application takes a user query and uses the openAI backend (your gptAssistant) to build a response.
- **Report** (PDF File):
Describe the architecture, the concepts you used, the features you implemented, and whatever you consider interesting. Make a few screenshots that clarify the usage of your client. Document both, the backend and the client.
 - write in German or English
 - Have a title page with all relevant information (team members, title, ...)
 - Add screenshots of your client application to show how it's performing.
 - Use an appropriate diagram to visualize the architecture of your application.
 - Do not copy large code-fragments into the main report. But show at least one (interesting/relevant) code fragment and explain what it does.
 - Structure your report. Write at least four sections:
 - **Introduction** where you give some context and explain the goal of the project.
 - Documentation of the **Client**: What your kind of service your client is providing. Screenshots of the working client

- Some details about the openAI **Backend**. If you use File Search, Code Interpreter or Functions (or another external API), make sure to document it.
- **Discussion** of results and learnings
- If you have additional material which doesn't fit into the main report, but you don't want to delete, then add an appendix (a chapter at the end of the report). Typically, in the main text you would write something like "for technical details see appendix xy". Note that the appendix is optional; assume that we do **not** read it. Grading is based on the main text.
- Focus on relevant content. Be concise. Remove clutter:
<https://www.acrolinx.com/blog/how-to-tidy-up-your-writing/>

3.2. How to get extra points

We do not specify a detailed, exhaustive list of criteria to get extra points. Anything that goes beyond the minimal requirements is appreciated, for example:

- creativity/originality/depth of the ideas
- clarity and completeness of the report
- interesting use-case
- plausibility of the ideation and evaluation
- powerful usage/combination of the capabilities of assistants
- integration of an additional API/service
- development of a streamlined UI
- ...

3.3. How to get minus points

- paragraphs filled with empty chatGPT blabla.
- superficial ideation (3 weak ideas with a pro-forma evaluation)
- non-professional system documentation (lack of clear diagrams)
- figures without caption
- ...

4. How to submit your report

There will be an assignment called "Graded Project Report Submission" on Moodle (in the week 7 section). **All team members will have to upload the (same) PDF File.** Do not upload code.

5. Manage your resources!

It is easy to get lost in the sea of possibilities and spend way too much time on this project. This project is worth 40% of your final AIFo-grade. Manage your resources accordingly.

6. Respect the deadline

If you do not hand-in a report by the deadline, we have to give you a grade 1.0. If you are unable to meet the deadline for important reason (illness, military service, etc.) the deadline can be discussed. Contact us as early as possible!